

JEE MAIN REVISION MASTER: PARABOLA 2026 0 QUESTIONS 0% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Kcet Parabola - Core Application	3 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Parabola - Core Application (Moderate Yield — 3 PYQs)

Core Principle: Key problem pattern in Kcet Parabola.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "The equation of parabola whose focus is (6,0) and directrix is $x=-6$ is..." [Ans: N/A]
- [2021 Shift] Q#2 (MCQ): "If the parabola $y=\alpha x^2-6x+\beta$ passes through the point (0,2) and has its tangent at $x=(3/2)$ parallel to X-axis, the..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "If the parabola $x^2=4ay$ passes through the point (2,1), then the length of the latus rectum is..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

The equation of parabola whose focus is (6,0) and directrix is $x=-6$ is

- (A) $y^2=24x$
- (B) $y^2=-24x$
- (C) $x^2=24y$
- (D) $x^2=-24y$

Correct Answer: None

Solution Analysis: ■ Focus has positive x-coordinate. So, coordinate of focus $= (a,0) \equiv (6,0) \therefore a=6$ (i) Hence, equation of parabola is $y^2=24x$ [■ $y^2=4ax$ and using Eq. (i)]